

Workflows and System Architecture Workstream

Introductions

Peter Groszkowski

ORNL

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on behalf of the openQSE Workflows and System Architecture Working Group



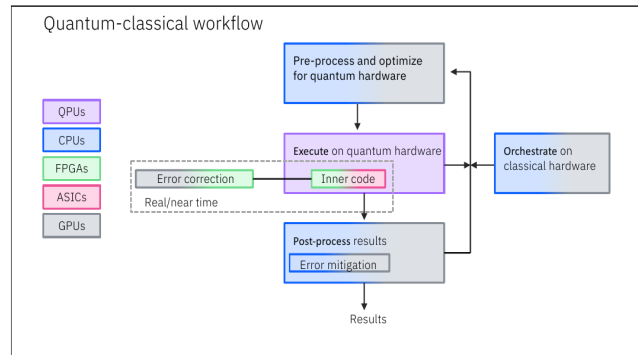
openQSE Workflows and System Architecture WG

Lead by Johannes Blaschke, Ellison Institute of Technology Oxford

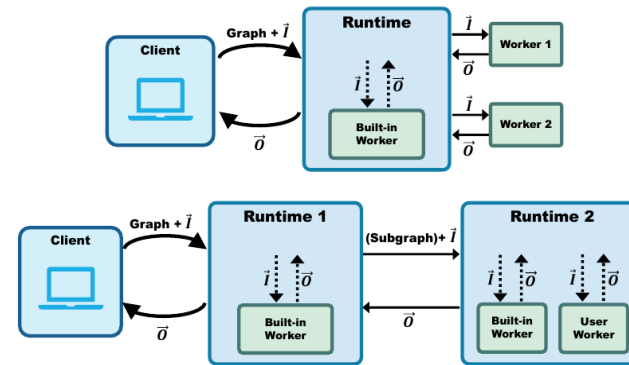
- **Application workflow pattern analysis**
 - Examining how target HPC-QC applications structure their computational workflows (classical pre-processing, quantum execution, post-processing, iteration loops)
- **Performance-related integration requirements**
 - Latency constraints, throughput demands, synchronization needs, and timing requirements that applications impose on the system architecture
- **Deriving system requirements**
 - Translating application-level needs into actionable input for an openQSE system architecture recommendation

Integration of quantum and HPC resources

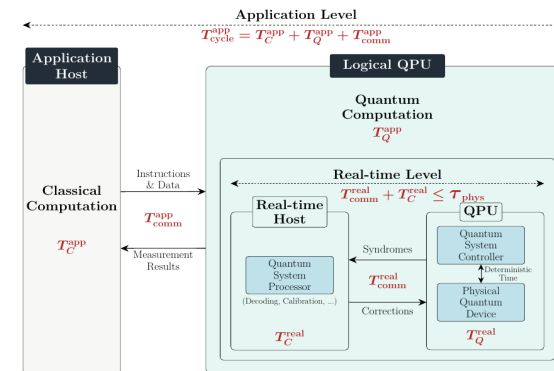
Timely topic of great interest to groups around the world!



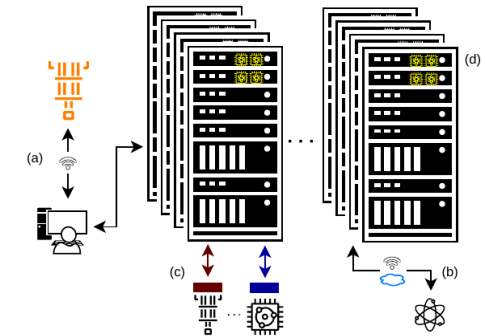
IBM: arxiv:2603.10970



Quantinuum: arxiv:2211.02350



NVIDIA+AWS: arxiv:2607.15426

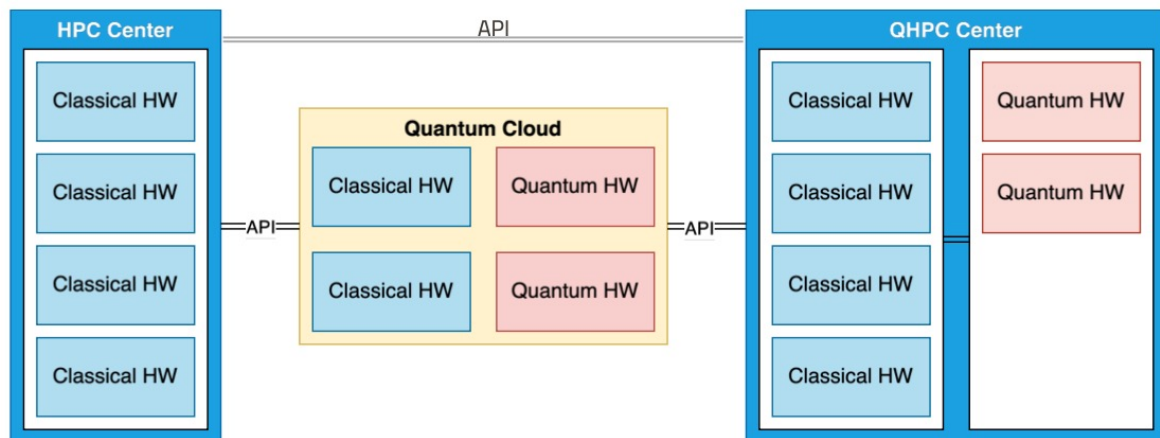


TUM: arxiv:2407.18527

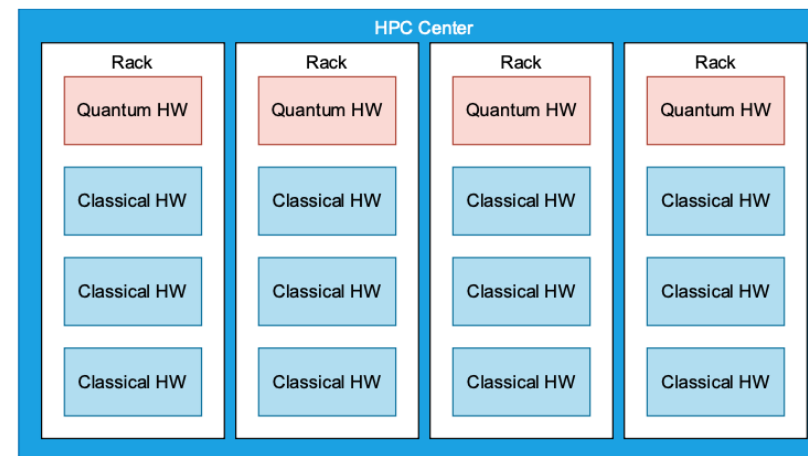
- Different institutions will have different requirements (and resources), but can we find commonalities?
- Shared knowledge can help with future interoperability and speed up development
 - Developing shared terminology and concepts
 - Identifying applications "primitives" and how they map to hardware usage

Integration Strategies

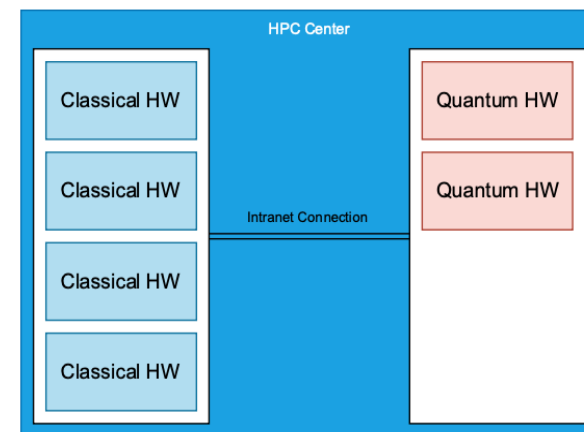
Y. Simmhan and A. R. Hatim, "Quantum-Classical Workflow Patterns", openQSE Seminar, April 4, 2026



Far-Quantum integration model (including cloud)



Co-located, tight integration



Co-located, loose integration

Classical Management of Quantum Resources
No awareness of quantum information flow

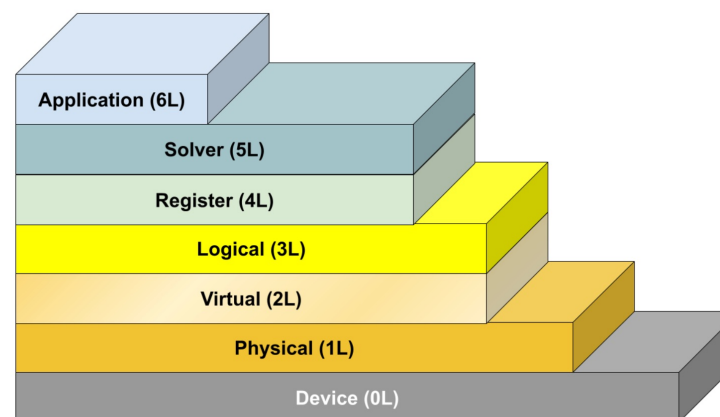
Classical Invocation of Quantum Solvers
Classical management of quantum information flow

Hybrid Implementation of Quantum Solvers
Logical quantum operations + classical control

Expression of fault-tolerant quantum algorithms
The quantum computing primitives

Hybrid Implementation of Logical Quantum operations
Physical quantum operations* + classical control

Expression of Physical quantum algorithms
Based on controllable quantum system of device



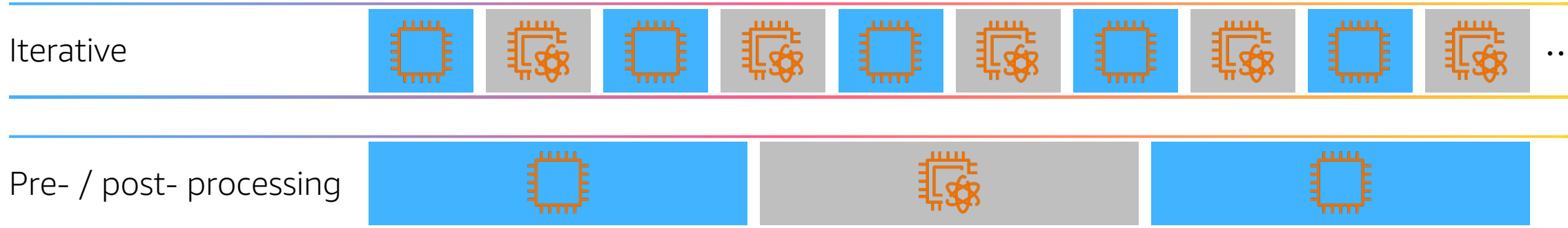
K. D. Kissel, "Hybrid Quantum HPC", openQSE Seminar, March 30, 2026

From Workflows to System Requirements

Workflow Patterns describe **structure** in the **data and control flow** of a hybrid quantum-classical application.

Integration Patterns describe how **workflow patterns** are mapped to classical and quantum infrastructure, and how **components are deployed and coupled**.

Y. Simmhan and A. R. Hatim, "Quantum-Classical Workflow Patterns", openQSE Seminar, April 4, 2026



How to reason about integration requirements?

A Performance Model for Hybrid Quantum-Classical Workflows

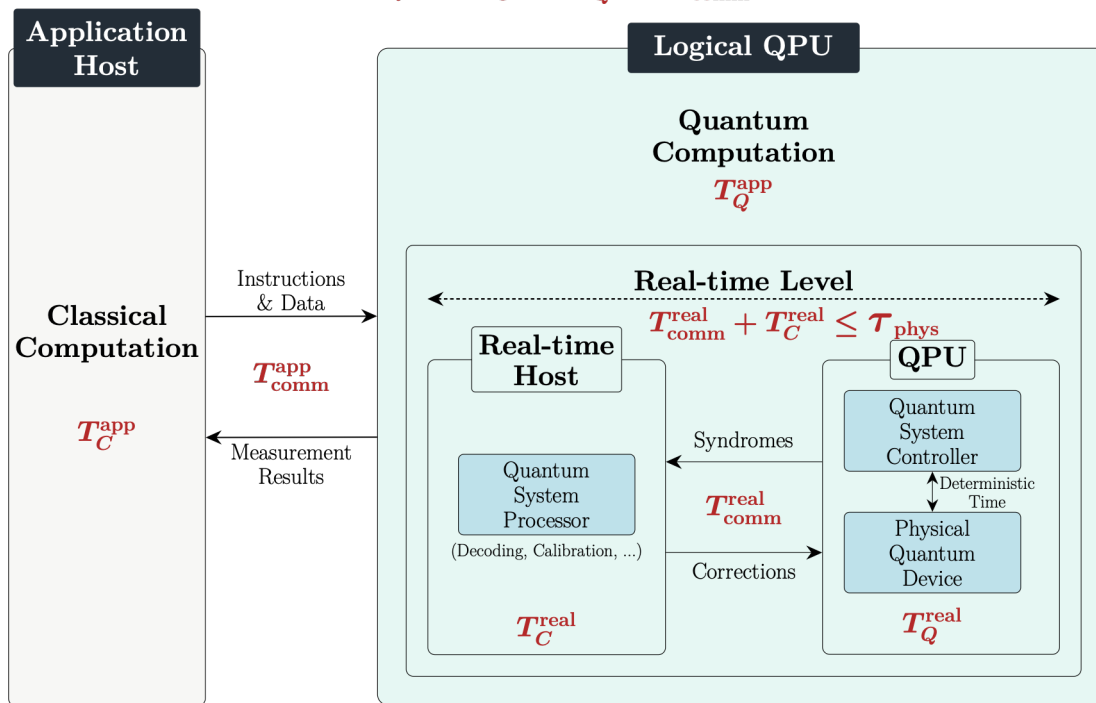
arXiv:2607.15426

Compute Cycle

A round trip between classical and quantum resources

Application Level

$$T_{\text{cycle}}^{\text{app}} = T_C^{\text{app}} + T_Q^{\text{app}} + T_{\text{comm}}^{\text{app}}$$



Runtime model to decompose workflow into classical- and quantum-computing and communication costs

Application level: Communication-to-computation ratio to diagnose integration requirements

Real-time level: Feasibility constraint to determine if execution is possible

Key insights:

- Co-location offers negligible performance benefit to compute-intensive applications
- Tight integration remains crucial for real-time tasks like QEC



Rao, Trenev, Gonthier, Patti, Stern, Takeshita, McArdle, Lietz, Klymko, Rrapaj, Tubman, Svore, Komar, Kyoseva

Agenda for the Crosstalk Sessions

Morning Session 10:00 a.m. – 12:30 p.m.		
Introductions	Peter Groszkowski, Sebastian Stern	15 mins
Lightning Talk: ORNL system integration efforts	Chris Zimmer (ORNL)	10 mins
Whole-room discussion		20 mins
Lightning Talk: SQD	Smriti Bajaj (Dell)	10 mins
Whole-room discussion		20 mins
Lightning Talk: Circuit cutting	Namit Anand (HPE)	10 mins
Whole-room discussion		20 mins
Lightning Talk: GQE	Tyler Takeshita (AWS)	10 mins
Whole-room discussion		20 mins

Mods: Sebastian Stern, Kevin Kissell

Afternoon Session 2:15 p.m. – 4:15 p.m.		
Lightning Talk: System integration for QEC	Peter Groszkowski (ORNL)	10 mins
Whole-room discussion		20 mins
Non-performance related requirements discussion	Laura Schulz (Argonne)	30 mins
General discussions:		30 mins
- Overlap with other working groups - Future milestones and deliverables		
Synthesize report-out for openQSE plenary		30 mins

Mods: Tyler Takeshita, Laura Schulz

Any volunteers for helping to take notes?

Position statements – Challenge or Support

1. No hybrid workflows we know today actually need HPC
2. No end-user application requires tight integration
3. For quantum-classical systems, tight integration does not require co-location